

RANDOM PRODUCT-PLUS-ONE COALESCENCE: EXACT ANTICHAIN TRANSFORMS AND MOMENT RADII

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ABSTRACT. Start with n ordered copies of 1. At each step choose a current adjacent pair uniformly and replace (x, y) by $xy + 1$; write X_n for the terminal value. We identify the deletion order of the original boundaries as a uniform permutation, which gives the random-binary-search-tree split law $X_n \stackrel{d}{=} 1 + X_I X'_{n-I}$. For this specific parameter we derive the complete finite-law recursion, a triangular differential hierarchy for all raw moments, and a closed bivariate transform for the expected cardinality-marked antichain polynomial of the induced tree. The mean has an elementary tangent form and satisfies

$$\mathbb{E}X_n \sim \rho^{-n}, \quad \rho = 1 - \exp\left(-\frac{2\pi}{3\sqrt{3}}\right),$$

with leading coefficient exactly one. For every integer $r \geq 1$, the r th-moment series has a normalized positive simple pole at a radius ρ_r , where $1 = \rho_0 > \rho_1 > \rho_2 > \dots > 0$, and $\limsup_n (\mathbb{E}X_n^r)^{1/n} = \rho_r^{-1}$. We do not claim a full coefficient asymptotic for $r \geq 2$. Finally, $X_n \geq n$ and $\mathbb{P}(X_n = n) = 2^{n-2}/(n-1)!$ for $n \geq 2$. Classical random-tree, forest-ideal, and analytic machinery is explicitly owner-subtracted.

1. INTRODUCTION AND SCOPE

Consider the following rank-decreasing random dynamics. Begin with the ordered word $(1, \dots, 1)$ of length n . If the current word has j entries, choose one of its $j-1$ adjacent pairs with probability $1/(j-1)$ and apply

$$(1) \quad (x, y) \mapsto xy + 1.$$

After $n-1$ mergers a positive integer X_n remains. The local rule is nonassociative, so X_n depends on the entire merger genealogy. The first nondegenerate law already occurs at $n=4$:

$$\mathbb{P}(X_4 = 4) = \frac{2}{3}, \quad \mathbb{P}(X_4 = 5) = \frac{1}{3}.$$

The genealogy is classical. Each current boundary is an undeleted original boundary, and successive uniform choices therefore form a uniform permutation. Its max-Cartesian tree has the usual random binary-search-tree (BST) root-split law. We give this identification zero contribution credit. The same subtraction applies to deterministic subtree enumeration [7], ideals of forest posets [5], antichain statistics on rooted plane trees [4], and random forest ideals [3]. Random-BST generating-function schemes and singularity methods are also established [1, 2, 6].

After those deductions, the paper records a narrow, parameter-specific exact package:

- (i) the full finite distribution recursion and the triangular hierarchy $F'_r = \sum_{k \leq r} \binom{r}{k} F_k^2$ for every raw moment;
- (ii) a closed bivariate ordinary generating function for expected antichains marked by cardinality;
- (iii) a strict cascade of moment radii, together with the exact elementary mean and its unit leading asymptotic coefficient; and
- (iv) the minimum output and its exact probability for every n .

The higher-moment conclusion is deliberately limited to a positive simple pole and an exact exponential *limsup*. Without control of all complex singularities, no full asymptotic is asserted for $r \geq 2$.

Two complementary viewpoints organize the proofs. The temporal viewpoint uses uniform deletion orders, conditional root splits, moments, and Riccati–Euler equations. The structural viewpoint regards the same orders as heap labelings of Cartesian trees and interprets the output as a cardinality-marked antichain count. Both viewpoints use the same split law; we do not present them as probabilistically independent proofs. The second viewpoint supplies a coefficient interpretation and exact comb controls rather than a new random-tree model. The bounded owner search behind this scope statement is a no-hit audit, not evidence of novelty or priority. External circulation remains HOLD.

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2. UNIFORM BOUNDARY DELETION AND THE FINITE LAW

Label the $n - 1$ original boundaries by $1, \dots, n - 1$. A merger deletes exactly the boundary separating the two merged blocks; every other current boundary retains its original label.

Theorem 2.1 (Split law and exact finite distribution). *Let I_n be uniform on $\{1, \dots, n - 1\}$. Conditional on $I_n = i$, let X_i and X'_{n-i} be independent copies generated on the two sides of the last-deleted boundary. Then*

$$(2) \quad X_1 = 1, \quad X_n \stackrel{d}{=} 1 + X_{I_n} X'_{n-I_n} \quad (n \geq 2).$$

Consequently, if $p_n(v) = \mathbb{P}(X_n = v)$, then $p_1(v) = \mathbf{1}_{\{v=1\}}$ and

$$(3) \quad p_n(v) = \frac{1}{n-1} \sum_{i=1}^{n-1} \sum_{\substack{a, b \geq 1 \\ 1+ab=v}} p_i(a) p_{n-i}(b).$$

Every law in (3) has finite support.

Proof. When j blocks remain, precisely $j - 1$ original boundaries remain, and the process selects each with probability $1/(j - 1)$. Thus the complete deletion order is a uniform permutation of $\{1, \dots, n - 1\}$. Its last element is uniform. Conditional on that element being i , the relative orders of the $i - 1$ left boundaries and the $n - i - 1$ right boundaries are independent and uniform. All mergers on each side occur before the last deletion, after which (1) combines the two terminal side values. This proves (2). Conditioning on I_n and the two side values gives (3). Finiteness follows inductively. $\square$

The proof also fixes the orientation: the *last* deleted boundary is the root of the evaluation tree. Reversing the deletion order exchanges maxima and minima in the Cartesian-tree description and is not the process above.

3. HEAP LABELINGS AND A MARKED ANTICHAIN TRANSFORM

For a fixed deletion order, let T be the ordered full binary evaluation tree. Its leaves carry 1, and every internal vertex applies $V(T) = 1 + V(T_L)V(T_R)$. Order the internal vertices by ancestry. For an indeterminate s , define

$$P_T(s) = \sum_B s^{|B|},$$

where B ranges over antichains of internal vertices, including the empty antichain. Put $P_{\text{leaf}}(s) = 1$.

Lemma 3.1 (Deterministic antichain identity). *For every evaluation tree,*

$$(4) \quad P_T(s) = s + P_{T_L}(s)P_{T_R}(s), \quad V(T) = P_T(1).$$

Proof. An antichain either contains the root, in which case it is the singleton root, or excludes the root, in which case its restrictions to the two subtrees are arbitrary antichains and can be chosen independently. The first identity follows. Its specialization at $s = 1$ is the same recursion and initial condition as $V(T)$. $\square$

Let T_n denote the tree induced by a uniform boundary order and write

$$a_n(s) = \mathbb{E}P_{T_n}(s), \quad A(z, s) = \sum_{n \geq 1} a_n(s) z^{n-1}.$$

Theorem 3.2 (Expected cardinality-marked antichains). *Coefficientwise as a formal series in z and s ,*

$$(5) \quad \partial_z A = A^2 + \frac{s}{(1-z)^2}, \quad A(0, s) = 1.$$

Set $w = 1 - z$,

$$\delta = \sqrt{1 - 4s}, \quad \beta_{\pm} = \frac{1 \pm \delta}{2},$$

and

$$(6) \quad Y(w, s) = \frac{\beta_+ w^{\beta_+} - \beta_- w^{\beta_-}}{\delta}.$$

With the removable value

$$Y(w, 1/4) = w^{1/2} \left(1 + \frac{1}{2} \log w \right),$$

the solution is

$$(7) \quad A(z, s) = \frac{Y_w(1 - z, s)}{Y(1 - z, s)}.$$

In particular, $[z^{n-1}s^k]A$ is the exact expected number of k -element internal-node antichains, and $A(z, 1)$ is the mean series of X_n .

Proof. The split law and [lemma 3.1](#) give, for $n \geq 2$,

$$(8) \quad a_n(s) = s + \frac{1}{n-1} \sum_{i=1}^{n-1} a_i(s) a_{n-i}(s).$$

Multiplying by $(n-1)z^{n-2}$ and summing gives (5). The function in (6) satisfies

$$Y_{ww} + \frac{s}{w^2} Y = 0, \quad Y(1, s) = Y_w(1, s) = 1.$$

Because $\partial_z = -\partial_w$, the logarithmic derivative $A = Y_w/Y$ satisfies (5) and the initial condition. Uniqueness of formal solutions proves (7). Taking the limit $\delta \rightarrow 0$ in (6) gives the displayed removable value. The coefficient interpretation is the definition of $a_n(s)$, and [lemma 3.1](#) gives $a_n(1) = \mathbb{E}X_n$. $\square$

Equation (7) is interpreted formally near $s = 0$, or analytically with consistent branches. At $s = 0$ it gives $A(z, 0) = (1 - z)^{-1}$, recording the unique empty antichain. The marked transform adds a coefficient-level statement, but its derivation still uses the same classical root split as [theorem 2.1](#).

4. ALL RAW MOMENTS AND A STRICT RADIUS CASCADE

For integers $r \geq 0$ set

$$m_{r,n} = \mathbb{E}X_n^r, \quad F_r(z) = \sum_{n \geq 1} m_{r,n} z^{n-1}.$$

Thus $m_{0,n} = 1$ and $F_0(z) = (1 - z)^{-1}$.

Proposition 4.1 (Triangular moment hierarchy). *For $n \geq 2$ and $r \geq 0$,*

$$(9) \quad (n-1)m_{r,n} = \sum_{i=1}^{n-1} \sum_{k=0}^r \binom{r}{k} m_{k,i} m_{k,n-i}.$$

Equivalently,

$$(10) \quad F_r'(z) = \sum_{k=0}^r \binom{r}{k} F_k(z)^2, \quad F_r(0) = 1.$$

At level r , the only new function on the right is F_r^2 .

Proof. Condition on the split in (2), expand $(1+ab)^r = \sum_{k=0}^r \binom{r}{k} a^k b^k$, and use conditional independence. This gives (9). Coefficient extraction from the square of each F_k gives (10). $\square$

The hierarchy is closed successively in r , although it is not claimed to have an elementary solution for every r .

Theorem 4.2 (Strict moment-radius cascade). *There are numbers*

$$(11) \quad 1 = \rho_0 > \rho_1 > \rho_2 > \cdots > 0$$

such that ρ_r is the radius of convergence of F_r . For every $r \geq 1$,

$$(12) \quad F_r(z) = \frac{1}{\rho_r - z} + O(1) \quad (z \rightarrow \rho_r),$$

and

$$(13) \quad \limsup_{n \rightarrow \infty} (\mathbb{E}X_n^r)^{1/n} = \rho_r^{-1}.$$

Proof. Proceed by induction from $F_0(z) = (1 - z)^{-1}$ and $\rho_0 = 1$. For $r \geq 1$, separate the new unknown in (10):

$$(14) \quad F_r' = F_r^2 + G_r, \quad G_r = \sum_{k=0}^{r-1} \binom{r}{k} F_k^2.$$

On $|z| < \rho_{r-1}$, let U_r solve

$$(15) \quad U_r'' + G_r U_r = 0, \quad U_r(0) = 1, \quad U_r'(0) = -1.$$

The logarithmic derivative $-U'_r/U_r$ satisfies (14) and agrees with F_r near zero.

By induction, all terms in G_r except the one containing F_{r-1} are analytic at ρ_{r-1} , while (12) gives

$$(16) \quad G_r(x) = \frac{r}{(\rho_{r-1} - x)^2} + O\left(\frac{1}{\rho_{r-1} - x}\right) \quad (x \uparrow \rho_{r-1}).$$

Choose c with $1/4 < c < r$. Close enough to ρ_{r-1} , $G_r(x) \geq c(\rho_{r-1} - x)^{-2}$. Every nonzero solution of the Euler comparison equation

$$V'' + \frac{c}{(\rho_{r-1} - x)^2} V = 0$$

has infinitely many zeros accumulating at ρ_{r-1} : after putting $s = \rho_{r-1} - x$, it is a linear combination of $s^{1/2} \cos(\nu \log s)$ and $s^{1/2} \sin(\nu \log s)$, with $\nu = \sqrt{4c - 1}/2$. Sturm comparison therefore forces U_r to vanish strictly before ρ_{r-1} . Let ρ_r be its first positive zero. Then $0 < \rho_r < \rho_{r-1}$.

The coefficient G_r is analytic at ρ_r . A double zero of U_r would contradict uniqueness for (15), so the zero is simple. Consequently $-U'_r/U_r$ has the normalized local form (12).

It remains to identify the positive pole with the complex radius. The coefficients of F_r are positive. If its radius R were smaller than ρ_r , Pringsheim's theorem would force a singularity at the positive point R [1, Section IV.6]. Yet G_r is analytic there and U_r has no positive zero before ρ_r , so $-U'_r/U_r$ is analytic in a neighborhood of R , a contradiction. Thus $R = \rho_r$. Cauchy–Hadamard now gives (13); replacing the exponent $1/(n-1)$ by $1/n$ does not change the limit superior. $\square$

Remark 4.3 (Hard ceiling for $r \geq 2$). The local pole in (12) lies at the positive convergence radius, but the proof does not exclude other singularities of the same modulus. Accordingly, [theorem 4.2](#) claims only (13) for $r \geq 2$. A statement such as $\mathbb{E}X_n^r \sim \rho_r^{-n}$ would require an additional dominant-singularity theorem and is not part of this paper.

5. THE CLOSED MEAN AND THE MINIMUM ATOM

At $r = 1$, the forcing in (14) is explicit. This permits a complete coefficient asymptotic, unlike the higher levels.

Theorem 5.1 (Elementary mean and unique dominant pole). *Let $M = F_1$ and $w = 1 - z$. Then*

$$(17) \quad M' = M^2 + \frac{1}{(1-z)^2}, \quad M(0) = 1,$$

and

$$(18) \quad M(z) = \frac{1}{w} \left[\frac{1}{2} - \frac{\sqrt{3}}{2} \tan\left(\frac{\sqrt{3}}{2} \log w - \frac{\pi}{6}\right) \right].$$

Its unique dominant singularity is the simple pole

$$(19) \quad \rho = 1 - \exp\left(-\frac{2\pi}{3\sqrt{3}}\right) = 0.7015639408\dots$$

For some $R > \rho$,

$$(20) \quad \mathbb{E}X_n = \rho^{-n} + O(R^{-n}), \quad \rho^{-1} = 1.4253868276\dots$$

In particular, the leading coefficient is one.

Proof. Equation (17) is the case $r = 1$ of [proposition 4.1](#). Write $M = -U'/U$. Then

$$U'' + \frac{U}{(1-z)^2} = 0, \quad U(0) = 1, \quad U'(0) = -1.$$

With $\alpha = \sqrt{3}/2$, the solution is

$$(21) \quad U(z) = \frac{2}{\sqrt{3}} w^{1/2} \cos\left(\alpha \log w - \frac{\pi}{6}\right).$$

Since $M = U_w/U$, differentiation yields (18).

In $|z| < 1$, the number $w = 1 - z$ lies in the right half-plane, so the principal logarithm is analytic. The zeros of the cosine in (21) satisfy

$$w_k = \exp\left[\frac{2}{\sqrt{3}} \left(\frac{2\pi}{3} + k\pi\right)\right], \quad k \in \mathbb{Z}.$$

They are positive real because all complex zeros of cosine are real. The zero $k = -1$ gives (19). The zeros with $k \leq -2$ correspond to larger positive $z < 1$, while those with $k \geq 0$ give $|z| > 1$; the logarithm first branches at $z = 1$. Hence ρ is the unique singularity on its circle and is isolated from the next one.

The logarithmic derivative of a simple zero has local form $1/(\rho - z) + O(1)$. Coefficient extraction gives (20). $\square$

The opposite edge of the distribution also has a closed form.

Proposition 5.2 (Minimum output and comb probability). *For every $n \geq 1$, $X_n \geq n$. For $n \geq 2$,*

$$(22) \quad \min \text{supp } X_n = n, \quad \mathbb{P}(X_n = n) = \frac{2^{n-2}}{(n-1)!}.$$

Proof. For positive integers, $xy + 1 \geq x + y$, with equality exactly when $x = 1$ or $y = 1$. Induct on n in (2). If the root split has sizes i and $j = n - i$, then

$$1 + X_i X'_j \geq 1 + ij = i + j + (i-1)(j-1) \geq n.$$

Equality requires minimal values in both subtrees and an endpoint split $i = 1$ or $j = 1$. Iterating this condition gives precisely the planar combs. Among the $(n-1)!$ boundary orders, their number c_n satisfies $c_2 = 1$ and $c_n = 2c_{n-1}$ for $n \geq 3$. Thus $c_n = 2^{n-2}$, which proves (22). $\square$

6. EXACT COEFFICIENT CONTROLS AND PROOF DEPENDENCIES

The first marked polynomials make the coefficient interpretation concrete. They were obtained both from (8) and by averaging over every boundary order through $n = 9$.

TABLE 1. Exact expected antichain polynomials. The value at $s = 1$ is $\mathbb{E}X_n$. The machine-readable artifact continues through $n = 12$.

n	$a_n(s) = \mathbb{E}P_{T_n}(s)$	$\mathbb{E}X_n$
1	1	1
2	$1 + s$	2
3	$1 + 2s$	3
4	$1 + 3s + \frac{1}{3}s^2$	$\frac{13}{3}$
5	$1 + 4s + \frac{7}{6}s^2$	$\frac{37}{6}$
6	$1 + 5s + \frac{13}{5}s^2 + \frac{2}{15}s^3$	$\frac{131}{15}$
7	$1 + 6s + \frac{47}{10}s^2 + \frac{59}{90}s^3$	$\frac{556}{45}$
8	$1 + 7s + \frac{263}{35}s^2 + \frac{121}{63}s^3 + \frac{17}{315}s^4$	$\frac{787}{45}$

The standard-library verifier implements two independent finite calculations where such independence is literal: direct adjacent merging for each boundary permutation and dynamic programming from the split law. It also compares the marked tree polynomials with (5), raw moments with (10), the Euler-series logarithmic derivative with the mean coefficients, and the comb count with (22). The exact artifact `code/marked_antichain_coefficients.tsv` records $a_n(s)$, the first two moments, minimum mass, and support size through $n = 12$.

These checks do not prove the singularity statements. Conversely, the analytic proof does not turn the heap-labeling viewpoint into a second probability law: both temporal and structural calculations inherit the same uniform split from theorem 2.1. The honest dependency chain is

$$\text{uniform boundaries} \implies \text{BST split} \implies \begin{cases} \text{finite law and moment Riccati hierarchy,} \\ \text{heap-labeled tree and marked antichains.} \end{cases}$$

The two branches cross-check finite coefficients and extremal combs, while the Sturm–Pringsheim argument is specific to the temporal moment branch.

7. OWNERSHIP BOUNDARY AND CONCLUSION

Uniform random-BST splitting, Cartesian trees, deterministic antichains and forest ideals, bivariate generating functions, Riccati linearization, Pringsheim’s theorem, and generic singularity extraction are background. The cited sources receive full credit for those objects and methods. The residual recorded here is only the displayed rule’s finite law, marked transform, triangular moment system, strict radius cascade, elementary mean, and comb atom as one exact package. We do not claim that all equivalent terminology or unpublished work has been exhausted.

The product-plus-one rule is unusually tractable because one deletion order simultaneously exposes a random-tree genealogy and a closed nonlinear evaluation. That interaction yields the exact transform and the strict moment-radius separation. The first moment admits a complete dominant-pole analysis;

higher moments stop at the exact limsup in [remark 4.3](#). Novelty, priority, and external dissemination remain HOLD pending independent review.

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